CampusERP

Complete Product Guide

Higher Education Cloud ERP — multi-tenant SaaS

Drivion Solutions Private Limited

Piloting at Swami Rama Himalayan University (SRHU), Dehradun, India

This guide explains the ENTIRE project:

All modules + how it works + login credentials + how to run

November 2026 — Build verified: 129 routes, 491 tests, all gates green

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1 About the Project

What CampusERP is, and why it exists

1.1 What it is

CampusERP is a next-generation, multi-tenant Higher Education ERP built by Drivion Solutions. It runs a university end-to-end on one platform: admissions, student lifecycle, fees, examinations and results, outcome-based education, accreditation reports (NAAC/NBA/NIRF), attendance, hostel, HR/payroll, and alumni — plus every minor day-to-day university function: notices, events, scholarships, health, library, transport, ID cards, parent portal, certificate generation, and more.

It is engineered specifically for Indian multi-disciplinary state and private universities. It is being piloted at Swami Rama Himalayan University (SRHU), Dehradun. The system replaces older, complex 2010s-era college software with a modern, fast, light-themed, fully-English interface that is still powerful enough for the registrar, exam cell, finance, HR, students, and parents.

One platform, every part of the university

From a student's first application to their alumni profile years later, every process lives in CampusERP. Central admin sees all schools; each school sees only its own data. Parents get a read-only view of their ward's progress.

1.2 The design principles

- One module at a time, built to a frozen spec before any code.
- Pure business-logic cores, each covered by automated tests (491 tests).
- Permission-based access — a user sees only what their role allows.
- Multi-tenant: every record is scoped to a schoolId.
- No fake/placeholder data in real flows; no dead buttons; no copied external product code.

2 The Complete Module Inventory

Everything the system contains at a glance

CampusERP currently ships 129 application routes across the following areas. Modules are grouped below by the implementation phase they were built in.

2.1 Core modules (foundation)

Area	What it manages
Admissions	Admission cycles, applications funnel, seat allocation, quotas, merit list
Student Lifecycle	Students, batches, enrollments, leaves, transfers, graduations, dossiers
Fees	Fee structures, invoices, payments, digital receipts
Examinations & Results	Exams, schedules, hall tickets, marks, SGPA/CGPA, grade cards
OBE (Outcome Based Education)	Program/course outcomes, CO-PO mapping, attainment
Accreditation	NAAC, NBA, NIRF report builders
Attendance	Period-wise marking, UGC 75% shortage detector
Hostel	Rooms, allocations, mess bills, complaints
HR & Payroll	Employees, leave, payroll, payslips
Alumni	Alumni profiles and network

2.2 Batch A — campus services

Area	What it manages
Library	Catalogue, circulation (issue/return), memberships, fines
Transport	Routes, vehicles, stops, transport passes with validity
Certificates	Official certificate generation and controlled issuance
Notices & Circulars	Create/publish notices; school/audience-scoped visibility

2.3 Batch B — student welfare & administration

Area	What it manages
Scholarships	Schemes, applications, review workflow, approvals
Grievance	Submit/track grievances, review and resolve
Health / Medical	Student health records and visits
Clubs & Events	Clubs, events, school-wide event management

2.4 Batch C — academic & exam enrichments

Area	What it manages

Area	What it manages
Course Registration	Course catalog, student registration, admin review
Grade Cards	Semester/annual grade cards with SGPA/CGPA and QR
Backlog & Revaluation	Backlog tracking, revaluation requests
Question Bank	Question bank management, paper-setting duty assignment

2.5 Batch D — career & alumni

Area	What it manages
Placement Cell	Companies, job posts, placements, student/admin views
Internships & Jobs	Listings, applications, student/admin views
Alumni Mentorship	Mentorship connections between alumni and students

2.6 Batch E — parent portal, reports, ID cards & UX

Area	What it manages
Parent Portal	Read-only ward dashboard: fees, attendance, results, progress
Reports Center	Aggregated CSV/PDF export hub across modules
ID Cards	Student identity cards with card number and QR payload
Cross-Module UX	Shared Loading/Error/Empty states, data-driven command palette

3 How the University Workflow Runs

The end-to-end journey CampusERP automates

CampusERP is organised around the natural lifecycle of a student and the university. This is the high-level flow the system is built to support:

1. Admission — an applicant is created in a cycle, documents are verified, a merit list is computed (12th %, NEET rank, CAT %ile as applicable), a seat is frozen with a token payment, and the candidate is enrolled.
2. Student lifecycle — the enrolled student is placed in a batch and program; enrollments for semesters are created; transfers, leaves, and graduations are tracked.
3. Fees — fee structures with components are defined per program/year; invoices are generated; payments are recorded and receipts issued.
4. Attendance — faculty mark attendance period-wise; the system continuously detects students below the UGC 75% threshold and supports alerts to parents.
5. Examinations & results — exams and schedules are set, hall tickets generated, marks entered; SGPA/CGPA are computed automatically and grade cards/marksheets are produced with verification QR codes.
6. OBE & accreditation — course outcomes and program outcomes are mapped (CO-PO), attainment is computed, feeding NAAC/NBA/NIRF reports.
7. Campus life — students use the library, transport, hostel, health, clubs/events, grievance redressal, and ID cards; parents follow progress through the parent portal.
8. Career & alumni — students are supported by the placement cell, internships/jobs, and mentorship; after graduation they remain connected through the alumni module.

The central roles (Registrar, Finance Officer, Exam Controller) supervise across the flow, while faculty and students act within their permitted areas.

4 Multi-School & Multi-Tenancy Model

How data is isolated and shared

A university is modelled as a central administration plus autonomous constituent schools. Every record belongs to a school (schoolId). A central-office user (school = "ALL") can work across schools; a school user only ever sees their own school's data.

Code	School	Kind
SST	School of Science & Technology	Engineering & technology (B.Tech CSE/ALML, BCA)
HIMS	Himalayan Institute of Medical Sciences	Medical (MBBS, MD, Nursing, Paramedical)
SRHU	Swami Rama Himalayan University (Central)	Central administration

Why this matters

The same installation can serve many universities or schools, each fully isolated. This is the core multi-tenant SaaS property of CampusERP.

5 Login Credentials

Demo users, password, and schools

The development database is seeded with the users below. Every seeded account uses the same password:

Password for all demo users

Password@2026

5.1 Main accounts

Email	Role	Who / what they can do
admin@srhu.edu.in	SUPER_ADMIN	System Administrator — bypasses all checks; creates roles, sets permissions, creates officers. The management account.
registrar@srhu.edu.in	REGISTRAR_ADMIN	Registrar (central) — academic administration, org structure, most modules.
fo@srhu.edu.in	FINANCE_OFFICER	Finance Officer — fees, invoices, payments, payroll.
coe@srhu.edu.in	EXAM_CONTROLLER	Controller of Examinations — exams, schedules, hall tickets, marks, results, grade cards.
director.sst@srhu.edu.in	DEAN (HOD)	Director of School of Science & Technology — department head, academic governance.
dean.hims@srhu.edu.in	DEAN (HOD)	Dean of Himalayan Institute of Medical Sciences — department head.
sandeep.cse@srhu.edu.in	FACULTY	Faculty (Computer Science & Engineering, SST).
sanjeev.path@srhu.edu.in	FACULTY	Faculty (Pathology, HIMS).
meena.kapoor@srhu.edu.in	PROGRAM_COORDINATOR	Faculty coordinating the B.Tech CSE program.
classcoord.cse@srhu.edu.in	CLASS_COORDINATOR	Faculty coordinating the B.Tech CSE class group A.
dept.secretary@srhu.edu.in	DEPARTMENT_SECRETARY	Department secretary (registrar-level clerk within SST).

Email	Role	Who / what they can do
aarav.sharma@sst.srhu.edu.in	STUDENT	Student — Aarav Sharma (B.Tech CSE, SST).
rohan.mbbs@srhu.edu.in	STUDENT	Student — Rohan Rawat (MBBS, HIMS).

5.2 Custom roles created by the SUPER_ADMIN seed

Role	Scope	Permissions
Hostel & Admissions Officer	Global	hostel.manage, admissions.manage
SST Academic Coordinator	SST school	attendance.mark, attendance.view_all, marks.enter_draft, calendar.manage

5.3 How to log in

- Start the app (see section 12) and open http://localhost:3000.
- Enter any email above with the password Password@2026.
- You land on a role-appropriate dashboard; the left sidebar shows only the modules your role can access.

6 Roles, Permissions & Access Control (RBAC)

How the system decides who can do what

Every action a user can perform is controlled by a permission string in area.action format (for example fees.manage or attendance.mark). Built-in roles bundle these permissions. The SUPER_ADMIN can create custom roles and assign any permissions from the catalog.

6.1 The built-in roles

Role	Typical user	What they can do
SUPER_ADMIN	System Administrator	Everything — roles, permissions, officers. Bypasses all checks.
REGISTRAR_ADMIN	Registrar	Academic administration, org structure, admissions oversight, most management.
FINANCE_OFFICER	Finance Officer	Fee structures, invoices, collections, payroll.
EXAM_CONTROLLER	Controller of Exams	Exams, schedules, hall tickets, marks, results, grade cards.
DEAN	Dean / Director	Department-level academic governance and oversight.
FACULTY	Teacher	Marks attendance, enters marks, views classes/timetable.
STUDENT	Student	Own data: fees, attendance, results, grade card, library, hostel, services.
PARENT	Parent/guardian	Read-only ward dashboard (Batch E).

6.2 Dynamic RBAC (custom roles)

- SUPER_ADMIN creates unlimited custom roles, picking permissions from a fixed, clean catalog.
- Roles can be global or school-scoped.
- Hierarchical positions (HOD, Program Coordinator, Class Coordinator, Department Secretary) refine scope within a department/class/program.

7 Module-by-Module Guide

Features, pages, and who uses each area

7.1 Core modules

Admissions (/admissions)

Run admission cycles; manage applications through a stage funnel (Applied !’ Docs Verified !’ Merit Listed !’ Seat Frozen !’ Enrolled); allocate seats and apply quotas. Merit is computed from 12th PCM%, NEET rank, or CAT%ile as configured.

Student Lifecycle (/students)

Complete student dossier: enrollment/roll number, school, batch, program, CGPA, contacts. Handles batches, enrollments, leaves, transfers, and graduations.

Fees (/fees)

Define fee structures with components (tuition, exam, hostel, mess, caution deposit). Generate invoices, record payments, and issue verified digital receipts with watermark, GST, and transaction details.

Examination & Result (/exams, /results, /gradecard)

Set exams and schedules, generate hall tickets, enter marks. SGPA/CGPA are computed automatically on the UGC 10-point relative grading scale; grade cards/marksheets carry verification QR codes.

OBE (/obe)

Program outcomes, course outcomes, CO-PO mapping, and attainment reports for accreditation.

Accreditation (/reports/naac, /reports/nba, /reports/nirf)

Self-study/accreditation report builders and NIRF data assembly across criteria.

Attendance (/attendance)

Period-wise marking plus an automated shortage detector for the mandatory UGC 75% threshold.

Hostel (/hostel)

Rooms (AC/Non-AC), allocations, mess bills, and complaints.

HR & Payroll (/hr)

Employees, leave, payroll runs, and payslips (including /hr/payslip).

Alumni (/alumni)

Alumni profiles and network management.

7.2 Batch A — campus services

Library (/library)

Catalogue (/library/catalogue), circulation issue/return (/library/circulation), memberships (/library/memberships), and fines (/library/fines).

Transport (/transport)

Routes (/transport/routes), vehicles (/transport/vehicles), and passes (/transport/passes) with validity status (PENDING/ACTIVE/EXPIRED).

Certificates (/certificates)

Generate official certificates (degree, provisional, character, etc.) with controlled issuance.

Notices & Circulars (/notices)

Create and publish notices/circulars with an admin area (/notices/admin); visibility filtered by school/audience.

7.3 Batch B — student welfare & administration

Scholarships (/scholarships)

Schemes (/scholarships/schemes), applications and review (/scholarships/review), and student view (/scholarships/my).

Grievance (/grievance)

Submit and track grievances with status progression; staff review and resolve.

Health / Medical (/health)

Student health records and visits (/health/visits).

Clubs & Events (/clubs, /events)

Clubs, events, and school-wide event management.

7.4 Batch C — academic & exam enrichments

Course Registration (/registration)

Course catalog (/registration/catalog), student registration (/registration/my), and admin review (/registration/admin).

Grade Cards (/gradecard)

Semester/annual grade cards with computed SGPA/CGPA and QR verification.

Backlog & Revaluation (/backlog)

Backlog tracking and revaluation requests.

Question Bank (/questionbank)

Question bank management and paper-setting duty assignment (/questionbank/duty).

7.5 Batch D — career & alumni

Placement Cell (/placement)

Companies (/placement/companies), job posts (/placement/posts), and placements. Student view at /placement/my, admin at /placement/admin.

Internships & Jobs (/internships)

Listings (/internships/listings), student view (/internships/my), and admin (/internships/admin).

Alumni Mentorship (/mentorship)

Mentorship connections between alumni and students.

7.6 Batch E — parent portal, reports, ID cards & UX

Parent Portal (/portal/parent)

A read-only dashboard for parents/guardians to view their ward(s): fees, attendance, results, academic progress. Admins link parents to students at /portal/parent/manage; per-student detail at /portal/parent/[studentId]. Adds the PARENT role.

Reports Center (/reports/center)

A central export hub aggregating data from fees, results, attendance, certificates, and admissions into downloadable CSV or PDF reports. Pure builders compute each report.

ID Cards (/idcards)

Student identity cards with generated card numbers and a scannable QR payload. Students view their own card at /idcards/my; staff manage at /idcards.

Cross-Module UX

Shared Loading/Error/Empty states across every screen and a data-driven command palette that indexes all modules and shows only permitted actions.

8 Using the Application

Navigation, shell, command palette, states

8.1 Navigation

- After login, the left sidebar lists the modules you can access, filtered by your role.
- Each module has its own sub-navigation (e.g. Library !’ Catalogue / Circulation / Memberships / Fines).
- The command palette (top bar / shortcut) lets you jump straight to a module or action, showing only what your role permits.

8.2 Forms & states

- Every form has validation, loading, empty, and error states, so you always know what is happening.
- When data is missing, a friendly empty state shows the next action.
- When a calculation or query fails, a clear error message appears so you can retry.

8.3 Dashboards

- Each role lands on a dashboard with role-relevant KPIs (Finance sees dues; Exam Controller sees exam status; students see their own snapshot).

8.4 Campus switching

- Central users (school = ALL) can view data across schools; school users are locked to their own school automatically.

9 Security & Data Protection

How data is kept safe

- Authentication: bcrypt password hashing + JWT stored in an HttpOnly cookie (auth_token). Passwords are never stored in plain text.
- Authorization: every action checks your effective permissions; SUPER_ADMIN bypasses all checks.
- Multi-tenancy: every query is scoped to your schoolId — a school can never see another school’s data.
- No secrets are hard-coded; configuration comes from environment variables (.env).
- No fake/placeholder data is wired into real flows, and no dead buttons are shipped.

10 Quality, Testing & Verification

How the project is kept green

The entire system is verified before any release with four automated gates:

```
npx tsc --noEmit     # type-check  !’ 0 errors
npx eslint           # lint         !’ 0 errors / 0 warnings
npm test             # tests        !’ 491/491 pass
npm run build        # production    !’ exit 0, 129 routes
```

- 491 automated tests cover the pure business-logic cores (fees, results, attendance, transport, permissions, command palette, and more).
- Independent reviewer + tester flow: the person who builds a module does not approve it.

11 Technology Stack & Codebase Layout

What the project is built with

11.1 Tech stack

Layer	Choice
Framework	Next.js 14 (App Router), TypeScript

Layer	Choice
Styling	Tailwind CSS — light, premium theme
Database / ORM	Prisma (SQLite for development, PostgreSQL for production)
Auth	bcrypt + JWT HttpOnly cookie
Tenancy	schoolId scoping via AsyncLocalStorage
Tests	Node built-in test runner (node:test) via scripts/test.cjs

11.2 Where things live

- src/app — pages/routes (one folder per module).
- src/lib — business-logic cores, schemas, and server clients (e.g. fee-core.ts, reports-center-core.ts, perms.ts).
- src/lib/__tests__ — automated unit tests.
- prisma/schema.prisma — database models; prisma/migrations — applied migrations.
- docs/specs — frozen specifications for every module.
- docs/SYSTEM-OVERVIEW.md — architecture summary.

12 Operational Notes

Setup, reset, and deployment

12.1 Start the app

```
cd drivion-saas
npm install
npx prisma migrate deploy
npx prisma db seed
npm run dev
```

Open http://localhost:3000.

12.2 One-click Windows launcher

```
Double-click:  drivion-saas\run_saas.bat
```

Starts the server and opens the browser automatically.

12.3 Reset the database (fresh demo)

```
npx prisma migrate reset --force
npx prisma db seed
```

12.4 Production

Use PostgreSQL (swap the DATABASE_URL), then run npm run build and npm run start. The application deploys to Vercel, AWS, or Railway.

This completes the guide

To use the system: start it, log in with one of the seeded accounts (password Password@2026), and use the left sidebar. Every module you are allowed to access appears automatically.